



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

FIRST SEMESTER 2026-27
COURSE HANDOUT

Date: 01/08/2026

In addition to part I (General Handout for all courses appended to the Timetable) this portion gives further specific details regarding the course.

Course No	CHE F313
Course Title	Separation Processes II
Instructor in Charge	PROF. SOMAK CHATTERJEE
Tutorial Instructors	T1: Prof. Muddu Shashank Venkat T2: Prof. Pradipta Chattopadhyay
Teaching Assistant(s)	Mr. Animikh Goswami, Mr. Anubhav Shukla
Lecture Schedule	T Th F 5 (1223)
Tutorial Schedule	M 1 T1 (1231) T2 (1232)
Contact Details of I/C	
Chamber	Room 1216-M, FD-I, Chemical Engineering Department
Email	somak.chatterjee@pilani.bits-pilani.ac.in
Office Phone	01596-25-5757

1. Course Description:

Special equilibrium-based separations like humidification and water cooling, drying of wet solids, adsorption, crystallization etc., Mechanical separations like filtration, centrifugation, froth floatation etc., Solid separations based on size reduction including sieving operations and related equipment like crushers, mills, pulverizers etc., special separation processes like ion-exchange, membranes, chromatography etc.

2. Scope and Objective of the Course:

This course gives the basic understanding of Mechanical Operations, which involves size reduction, size separation, filtration, etc. and certain Mass Transfer Operations like humidification, drying, adsorption, membrane separation, and crystallization. It covers the fundamentals, application of each of these unit operations and the equipment used is discussed in detail. The objective of the course is to orient the students to the industrial practices, which involve both mechanical operations and mass transfer operations, so that they can understand them better.

3. Text Books:

TB McCabe W. L., and Smith J. M., & Harriott P., Unit Operations of Chemical Engineering, 7th Ed., McGraw-Hill International Edition, 2006.



4. Reference Books:

- RB.1 Coulson J. M. & Richardson J. F., Chemical Engineering (Volumes 1-6), Pergamon Press, London, 1978 & 1997.
- RB.2 Ernest J. Henley, J. D. Seader, D. Keith Rope Separation Process Principles, 3rd Edition International Student Version, 2011.
- RB.3 Christie J. Geankoplis, Transport Processes and Separation Process Principles, 2003.
- RB.4 Don W. Green, Robert H. Perry, Marylee Z. Southard; Perry's Chemical Engineers' Handbook; 9th Edition.

5. Course Plan:

Module No.	Lecture (L) & Tutorial (T) Sessions	Reference	Learning Outcomes
M1 Properties & handling of particulate solids	L1. Introduction – Classification of Chemical Processes	TB: Ch. 28	- Understanding of the characteristics of masses of particulate solids and thereby designing processes and equipment for dealing with streams containing such solids. - Able to calculate the crushing efficiency required for a given feed using various laws. - Apply the concepts learned in experiments in the lab. - Able to select the suitable equipment for a given feed in the industrial setting.
	L2. Characterization of Solid Particles – sphericity, equivalent dia. and nominal dia.		
	L3. Mixed particle size and size analysis – Methods of analysis, Average particle size, Screen analysis		
	L4. Properties of Masses of Particles – Storage and conveying of solids		
	L5. Mixing of Solids – Measure of mixing performance		
	L6-7 Size Reduction – principles, crushing efficiency, mechanical efficiency. Rittinger's law, Kick's law and Bond's law		
	L8. Equipment for size reduction – Crushers, Grinders, Ultrafine Grinders, Cutting machines. Open and Closed-circuit operation		
	T1. & T2. Exercise problems on characterization and properties of particles		
M2 Mechanical Separations	L9. Screening – Screen efficiency	TB: Ch. 29	Understanding the methods of separating solid particles from gases or liquids, separating liquid drops from gases or other liquids, and separating one type or size of
	L10. Filtration – types, filter media, filter aids, cake filtration, constant pressure filtration		
	L11. Continuous filtration, constant rate filtration. Centrifugal filtration, washing rate		



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

	L12. Clarifying Filters, Crossflow Filters and Membrane Filters. Concentration polarization L13. Gravity Sedimentation Processes – Gravity classifiers, sorting classifiers L14. Clarifiers and Thickeners, Design principles of Clarifiers and Thickeners, Sedimentation zones in continuous thickeners L15-16. Centrifugal Sedimentation – Cyclones, Hydro clones, Centrifugal decanters, Nozzle-discharge centrifuge Sludge separators, Principle of centrifugal Sedimentation T3. & T4. Exercise problems on mechanical separations		solids from a mixture of particles. • Able to make calculations on filtrations. • Apply the concepts in the experiments in the lab. • Able to select the filtration technique required for the any industrial processes.
M3 Humidification Operations	L17. Definitions and Humidity chart L18. Wet-bulb Temperature and Adiabatic saturation temperature T5. Exercise problems on humidification operation	TB: Ch. 19	• Understanding the transfer of materials between a pure liquid phase and a fixed gas that is nearly insoluble in the liquid • Able to use the humidity chart and solve problems on humidification.
M4 Drying of Solids	L19-20. Introduction – Classification of dryers, Solids handling in dryers Principles of Drying – Temperature patterns in dryers, heat transfer in dryers, Heat transfer unit Phase equilibria, Basic definitions L21-22. Cross-Circulation Drying – Rate of drying, Drying curve, Calculation of drying time under constant drying conditions Through-Circulation Drying – Drying of suspended particles; Freeze Drying L23. Drying Equipment – Dryers for solids and pastes, Dryers for solutions and slurries L24. Selection of drying equipment – Thermal efficiency, General considerations; Evaporator dryers	TB: Ch. 24	• Understanding the principles and types of drying, calculation of heat duty and design of dryers, the drying curve and calculation of drying time, drying equipment. • Able to solve the problems on calculation of drying time using the experimental data.



M5 Adsorption and Fixed-Bed Separations	T6. Exercise problems on Drying	TB: Ch. 25	- Understanding the principle of adsorption, adsorption isotherms, breakthrough curves, scale-up of adsorbers, ion exchange and chromatography. - Able to draw breakthrough curves, calculate the unused bed length, utilization ratio, and scale up from the experimental data. - Identify and select the separation technique required for a given industrial feed.
	L25. Adsorption – Adsorbents and adsorption processes, Fixed-bed adsorbers		
	L26. Adsorption isotherms – Types of isotherms		
	L27-28. Principles of adsorption, Breakthrough curves, Scaleup		
	L29. Basic equations for adsorption, Solution to mass transfer equations		
	L30. Adsorber design, Continuous operations		
	L31. Ion Exchange – Equilibria, Mass transfer rates, Operations of ion exchangers		
M6 Membrane Separation Processes	L32. Chromatography – Gas-Liquid Chromatography	TB: Ch. 26	Understanding the separation of gas mixtures with porous or non-porous membranes, liquid mixtures using dialysis, extraction, pervaporation, reverse osmosis.
	T7 & T8. Exercise problems on adsorption and fixed-bed separation.		
	L33-34. Separation of Gases – Porous membranes, polymer membranes, product purity and yield		
M7 Crystallization	L35-36. Separation of Liquids – Dialysis, extraction, pervaporation, Reverse osmosis, Concentration polarization	TB: Ch. 27	- Understanding crystal geometry, solubility curve, yields, supersaturation, nucleation, crystal growth and the crystallization equipment. - Able to calculate the energy required for a given crystallization process.
	T9. Exercise problems on membrane separation processes		
	L37. Crystal Geometry		
	L38. Equilibria and Yields – Enthalpy balances, Supersaturation		
	L39. Nucleation and Crystal Growth		
	L40. Crystallization Equipment		
	T10. Exercise problems on crystallization		



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

Course Learning Outcomes:

- Select suitable equipment for size reduction, size separation, filtration, drying, crystallization and adsorption.
- Calculate average particle size of a mixture sample based on screen analysis.
- Design counter-current rotary dryers.
- Calculate the drying time for a batch of wet solids.
- Draw the breakthrough curve for an adsorption process using experimental data.
- Design the adsorber, calculate unused bed length, utilization.
- Identify suitable membrane processes for gas-gas and liquid-liquid separation.
- Understand the principles of gas and liquid chromatography.
- Understand the principle of crystal formation, its growth and yield calculation.

6. Evaluation Scheme:

Component	Duration (minutes)	Weightage (%)	Marks (300)	Date & Time	Nature of component
Mid Semester Test	90	30	90	06/10/2026 (FN1)	Closed Book
Comprehensive Examination	180	40	120	05/12/2026 (FN)	Closed and Open Book
Tutorial Tests (Best 3 of 6)	20	15	45 (Each tutorial test carries 15 marks)	3 before Midsem; 3 after Midsem (To be announced one week before, through notice, posted in Nalanda LMS)	Closed Book
Lecture Quizzes (Best 3 of 6)	20	10	30 (Each quiz carries 10 marks)	3 before Midsem; 3 after Midsem (Surprise in nature)	Closed Book
Assignment (2)	Two weeks	5	15 (Each assignment carries 7.5 marks)	1 before Midsem and 1 after Midsem	Take home type (Students can take the help of any reference material)

7. Chamber Consultation Hour:

- All Tuesdays: 4 pm to 6 pm (Room 1216-M, FD-I, Chemical Engineering Department). Prior appointment by email on any other days are also welcome.

8. Notices:

- All notices concerning this course will be displayed in Nalanda LMS Portal.

9. Make-up Policy:

- If the student is unable to appear for the Regular Test/Examination due to genuine exigencies, the student must refer to the procedure prescribed by AUGSD for applying for Make-up Test/Examination.
- Make-up will be granted only for extreme emergencies or for genuine reason(s).
- Proper proofs must be submitted along with the Make-up application.
- Prior permission of IC is compulsory.



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

-
- Decision of the IC in all matters of make-up shall be final.
 - No make-up will be granted for the tutorials/lecture quizzes/discussion/projects/assignments.

10. Criteria for NC (Not Cleared):

- In case application for the make-up is rejected due to the lack of genuineness, a student will be reported NC (both for Midsem and/or Comprehensive examination).
- If the student remains absent in all quizzes and/or tutorial tests and/or fails to submit all the assignments, then the student will be reported NC.

11. Note (if any):

- Any changes in the course handout will be communicated through course Nalanda LMS.

(PROF. SOMAK CHATTERJEE)
Instructor-in-charge
CHE-F313